

✓ SIMPLE REGRESSION ANALYSIS

Marketing ROI analysis using TV, Radio and Social Media as features. dependent variable/ target variable is Sales, while the independent variables/features are TV, Radio and Social Media. The goal of this project is to recommend the marketing channel with the strongest ROI impact

import libraries and packages

pandas for data manipulation

statsmodels for statistical modeling

seaborn for plotting

✓ Import packages

import pandas as pd
import numpy as np
import seaborn as sns

```
# Load dataset
marketing = pd.read_csv("marketing.csv")

# Examine first 5 rows of dataset
marketing.head(10)
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181
5	29.0	9.614382	1.027163	105.889148
6	55.0	24.893811	4.273602	198.679825
7	31.0	17.355042	2.289855	108.733932
8	76.0	24.648898	7.130116	270.189400
9	13.0	0.431128	2.229423	48.280582

```
#examine the shape of the data
marketing.shape
```

```
(4572, 4)
```

```
#examine the size of the data
marketing.size
```

```
18288
```

```
marketing.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0    TV          4562 non-null   float64
1    Radio       4568 non-null   float64
2    Social_Media 4566 non-null   float64
3    Sales       4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

```
marketing.describe()
```

	TV	Radio	Social_Media	Sales
count	4562.000000	4568.000000	4566.000000	4566.000000
mean	54.066857	18.160356	3.323956	192.466602
std	26.125054	9.676958	2.212670	93.133092
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.525957	1.527849	112.322882
50%	53.000000	17.859513	3.055565	189.231172
75%	77.000000	25.649730	4.807558	272.507922
max	100.000000	48.871161	13.981662	364.079751

```
marketing.columns
```

```
Index(['TV', 'Radio', 'Social_Media', 'Sales'], dtype='object')
```

DATA CLEANING

```
marketing.isnull().sum()
```

```
TV          10
Radio        4
Social_Media 6
Sales        6
dtype: int64
```

drop all missing values in datasets

```
marketing.dropna(inplace=True)
```

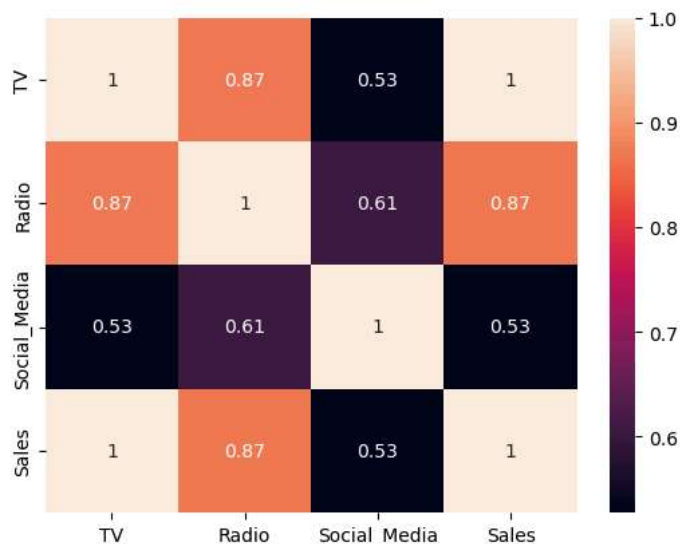
```
marketing.isnull().sum()
```

```
TV          0
Radio        0
Social_Media 0
Sales        0
dtype: int64
```

Exploratory data analysis

```
# create a heatmap of the dataset
sns.heatmap(marketing.corr(), annot=True)
```

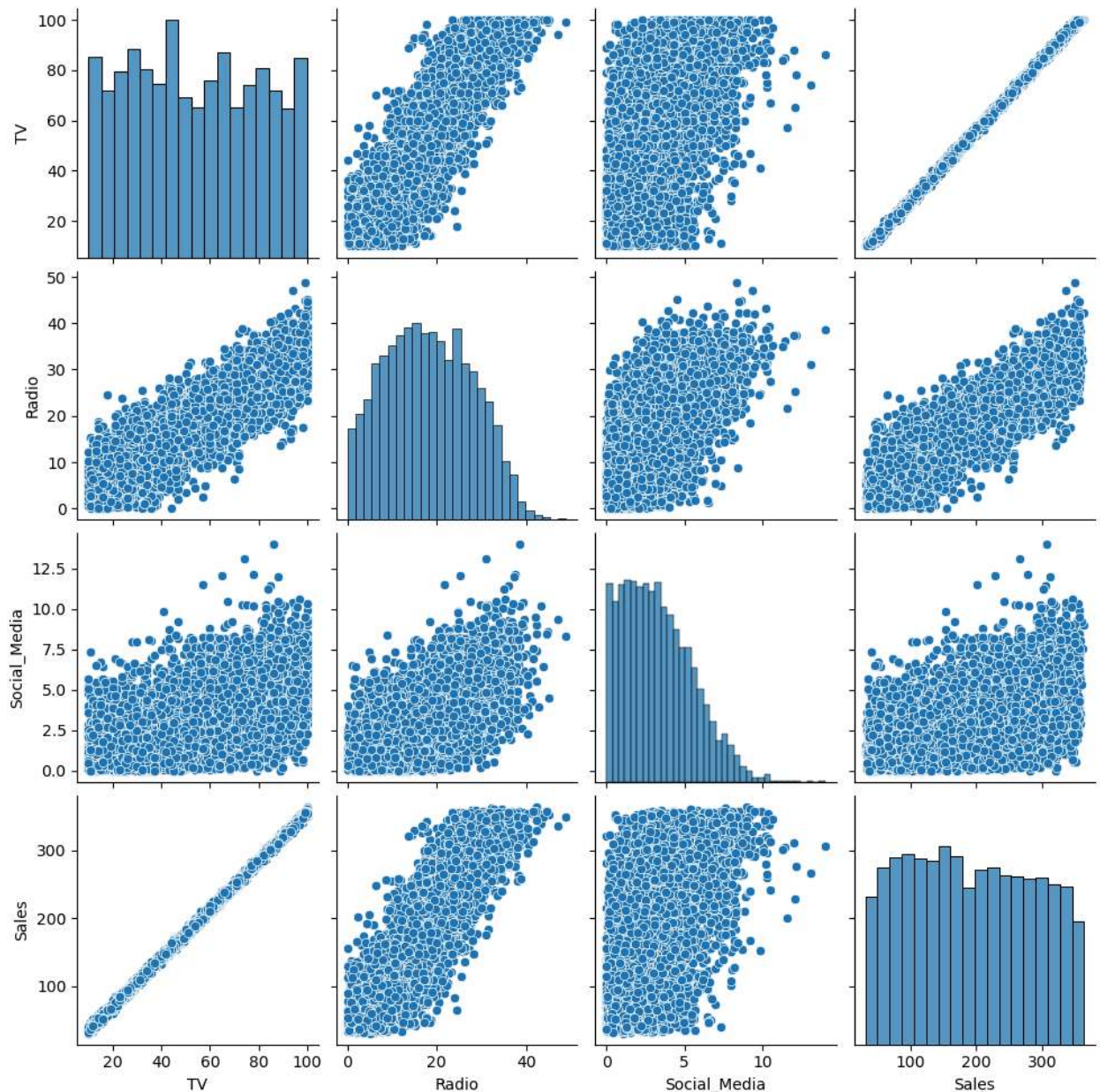
```
<Axes: >
```



from the heatmap This gives a visual explanation of the correlation between TV, Social_Media and Radio. TV has an almost perfect correlation with Sales TV has a higher impact on sales

```
# Create pairwise scatterplots of data set
sns.pairplot(marketing)
```

<seaborn.axisgrid.PairGrid at 0x1860faf12b0>



From the scatterplot matrix, you can observe a few linear relationships:

- TV and Sales (almost perfectly linear)
- TV and Radio
- Radio and Sales
- Social Media and Radio

Model development

Based on the above scatterplots, we could probably run a simple linear regression on any of the three relationships identified. For this project our focus will be on the relationship between TV and Sales as they are the most linearly correlated

To do this, we first subset the variables of interest from the dataframe by using double square brackets [[]], and listing the names of the columns of interest.

```
# Subset Data
```

```
# Subset Data

ols_data = marketing[['TV', 'Sales']]
```

Next, we can construct the linear regression formula

- Dependent variable y = Sales
- Independent Variable X = TV

```
# Write out formula
ols_formula = "Sales ~ TV"
```

$y = mx + c + e$

Let's build the simple linear regression model in statsmodels

```
# Import ols function
from statsmodels.formula.api import ols
```

```
# Build OLS, fit model to data
OLS = ols(ols_formula, ols_data)
model = OLS.fit()
```

Let's get the summary of the model, coefficients and more statistics about the model. this helps us evaluate the model and interpret the results

```
model.summary()
```

```

OLS Regression Results

Dep. Variable:  Sales      R-squared:  0.999
Model:  OLS      Adj. R-squared:  0.999
Method:  Least Squares      F-statistic:  4.517e+06
Date:  Thu, 11 Jun 2026      Prob (F-statistic):  0.00
Time:  09:18:42      Log-Likelihood:  -11366.
No. Observations:  4546      AIC:  2.274e+04
Df Residuals:  4544      BIC:  2.275e+04
Df Model:  1
Covariance Type:  nonrobust

   coef  std err   t    P>|t| [0.025 0.975]
Intercept -0.1325  0.101  -1.317  0.188 -0.330  0.065
TV      3.5615  0.002  2125.272  0.000  3.558  3.565

Omnibus:    0.052   Durbin-Watson:   1.998
Prob(Omnibus): 0.974   Jarque-Bera (JB): 0.031
Skew:       -0.001   Prob(JB):   0.985
Kurtosis:    3.012   Cond. No.   138.
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

FROM THE OLS SUMMARY

R-Squared: this indicates that TV explains 99% of Sales Variation

TV coef: 3.5615 what this means is that for every 1 unit increase in TV, Sales go up by 3.56 units (i.e 3.5X ROI)

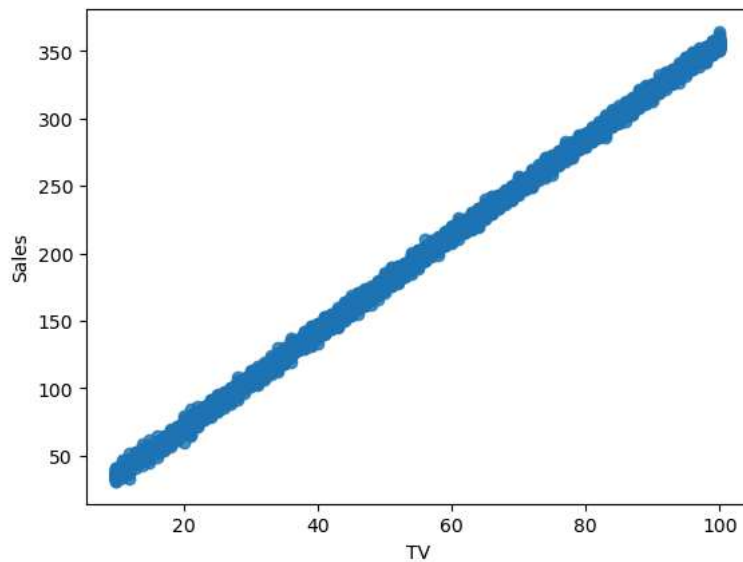
P|t| for TV is 0.00, so TV is statistically significant, as $p|t| < 0.00$ shows a feature is useful/ significant while > 0.00 signals noise

DF model = 1, tells us that this is a simple regression model as only one feature was used

visual of the regression line.

```
sns.regplot(x = "TV", y = "Sales", data = ols_data)
```

<Axes: xlabel='TV', ylabel='Sales'>



model assumptions

1. Linearity
2. Normality
3. Independent observations
4. Homoscedasticity

we already checked the linearity assumption by creating the scatterplot matrix and we observed that TV and Sales assume linearity

The normality and homoscedasticity assumptions focus on the distribution of errors. Thus, we can only check these assumptions after we have constructed the model. To check these assumptions, we will check the residuals, as an approximation of the errors.

To more easily check the model assumptions and create relevant visualizations.

```
X = ols_data["TV"]
```

```
X.head()
```

```
0    16.0
1    13.0
2    41.0
3    83.0
4    15.0
Name: TV, dtype: float64
```

```
# Get predictions from model
fitted_values = model.predict(X)
```

```
fitted_values
```

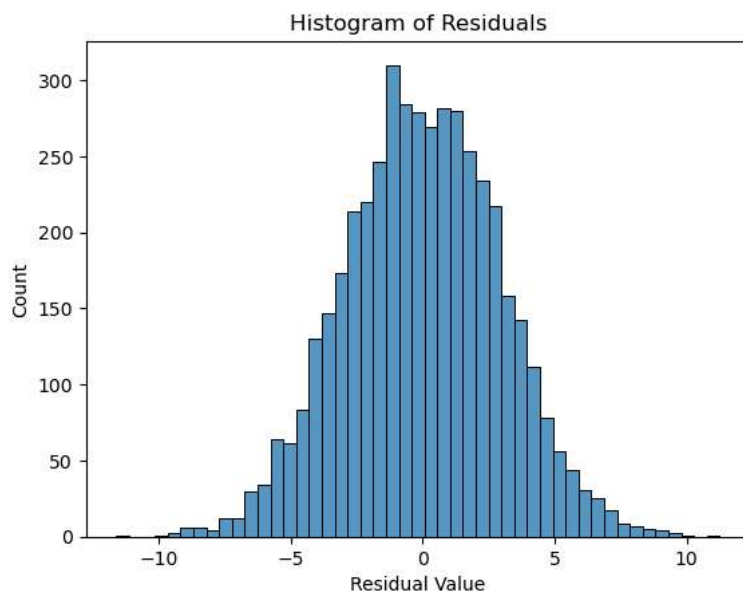
```
0      56.851733
1      46.167191
2     145.889585
3     295.473177
4      53.290219
...
4567    92.466874
4568   252.735008
4569   156.574128
4570   252.735008
4571   149.451099
Length: 4546, dtype: float64
```

```
# Calculate residuals
residuals = model.resid
```

Check the normality assumption

From the below histogram, you may notice that the residuals are almost normally distributed. In this case, it is likely close enough that the assumption is met.

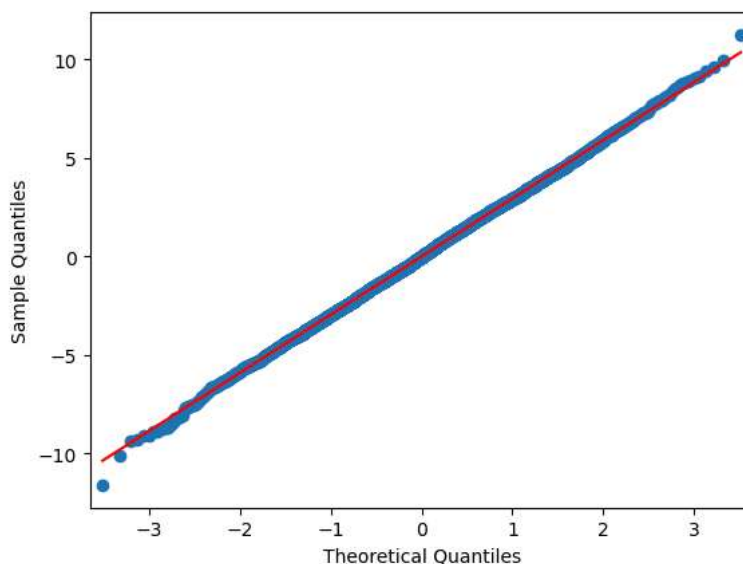
```
import matplotlib.pyplot as plt
fig = sns.histplot(residuals)
fig.set_xlabel("Residual Value")
fig.set_title("Histogram of Residuals")
plt.show()
```



Another way to check the normality function is to create a quantile-quantile or Q-Q plot. If the residuals are normally distributed, you would expect a straight diagonal line going from the bottom left to the upper right of the Q-Q plot.

The Q-Q plot shows a similar pattern to the histogram, where the residuals are mostly normally distributed, except at the ends of the distribution.

```
import matplotlib.pyplot as plt
import statsmodels.api as sm
fig = sm.qqplot(model.resid, line = 's')
plt.show()
```



✓ Check the homoscedasticity assumption

Lastly, we have to check the homoscedasticity assumption. To check the homoscedasticity assumption, we can create a scatterplot of the fitted values and residuals. If the plot resembles a random cloud (i.e., the residuals are scattered randomly), then the assumption is likely met.

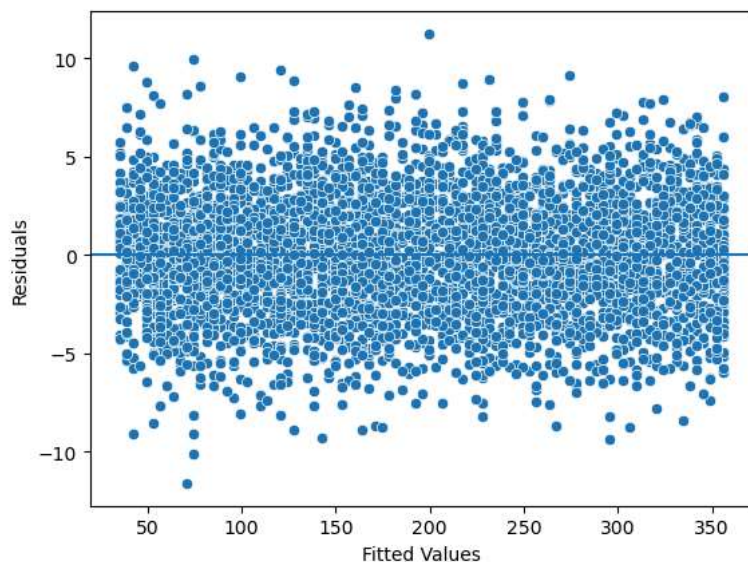
he first argument is the variable that goes on the x-axis. The second argument is the variable that goes on the y-axis.

```
import matplotlib.pyplot as plt
fig = sns.scatterplot(x=fitted_values, y=residuals)

# Add reference line at residuals = 0
fig.axhline(0)

# Set x-axis and y-axis labels
fig.set_xlabel("Fitted Values")
fig.set_ylabel("Residuals")

# Show the plot
plt.show()
```



✓ RECOMMENDATION

TV is the Dominant driver of Sales in this dataset with 3.523x ROI. The company should prioritize and increase TV budget allocation to maximize Sales revenue.

All OLS assumptions are met- Linearity, normality and homoscedasticity as confirmed by diagnostic plots

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